

Exponential and Logarithmic Equations

ANSWER KEY



Exponential equations

- 1 Solve $3^{x+1} = 81$ exactly.
 $3^{x+1} = 3^4$, so $x + 1 = 4$ and $x = 3$.
 - 2 Solve $5^{2x} = 17$. Give an exact answer and a decimal to three places.
 $2x = \log_5 17 = \frac{\ln 17}{\ln 5}$, so $x = \frac{\ln 17}{2 \ln 5} \approx 0.880$.
 - 3 Solve $e^{2x} - 5e^x + 6 = 0$ exactly.
Let $u = e^x$: $u^2 - 5u + 6 = (u - 2)(u - 3) = 0$, so $e^x = 2$ or $e^x = 3$: $x = \ln 2$ or $x = \ln 3$.
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Logarithmic equations

- 4 Solve $\ln(2x - 3) = 1$. Give an exact answer and a decimal to three places.
 $2x - 3 = e$, so $x = \frac{e+3}{2} \approx 2.859$.
- 5 Solve $\log_2 x + \log_2 (x - 6) = 4$, checking for extraneous solutions.
 $\log_2 (x(x - 6)) = 4$, so $x^2 - 6x - 16 = 0$, giving $(x - 8)(x + 2) = 0$. Reject $x = -2$ (logarithm of a negative number): $x = 8$.
- 6 Solve $\log(x + 3) - \log(x - 1) = 1$, checking for extraneous solutions.
 $\frac{x+3}{x-1} = 10$, so $x + 3 = 10x - 10$ and $x = \frac{13}{9}$. Check: $x > 1$, valid.
- 7 The pH of a solution is $\text{pH} = -\log[\text{H}^+]$. Find the hydrogen-ion concentration $[\text{H}^+]$ of a solution with pH 4.6, in scientific notation to two significant figures.
 $[\text{H}^+] = 10^{-4.6} \approx 2.5 \times 10^{-5} \text{ mol/L}$.